

Phase 2 automatic commissioning using auto scan

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Abstract

The ATLAS and CMS Phase 2 upgrade presents multiple challenges, one of which is the need to test the 2PACL based CO₂ detector cooling system’s performance of multiple units in a short time frame with a limited team. To address this requirement, various solutions were explored in the development of 2PACL control systems, ultimately leading to the implementation of a feature called Autoscan. This approach enables the automatic operation of different actuators, allowing them to follow multiple predefined test scenarios within a set time frame. Test requests are queued to facilitate continuous, large-scale testing, and the system can initiate tests outside of standard working hours, thereby extending the overall available testing time. The development process ensured that the Autoscan implementation remains independent of PLC development, making it easily adaptable to different subsystems without requiring tracking of multiple PLC code versions. In its current state, the feature fully leverages the UNICOS control framework, simulating manual operations from the SCADA interface to the PLC. The PLC retains its independent development strategy, unaffected by the Autoscan feature, and remains the primary system for interlocking in case of issues. Each subsystem hosts its own Autoscan functionality, ensuring that systems operate independently during scans. This feature allows cooling experts to focus on data analysis and provides a streamlined process for performing all necessary tests before the start of Long Shutdown 3 (LS3).

The problem we needed to solve

- Limited testing time due to restriction of CO₂ underground during LS3
- Small team dedicated to system testing
- Many system units to test with multiple “flavours” across 102 electrical cabinets

Plants:

- 1 x 1-head plant with 11 controllers, 15 actuators, 122 IOs
- 11 x 2-head plants with 15 controllers, 19 actuators, 146 IOs
- 4 x 3-head plants with 19 controllers, 23 actuators, 170 IOs

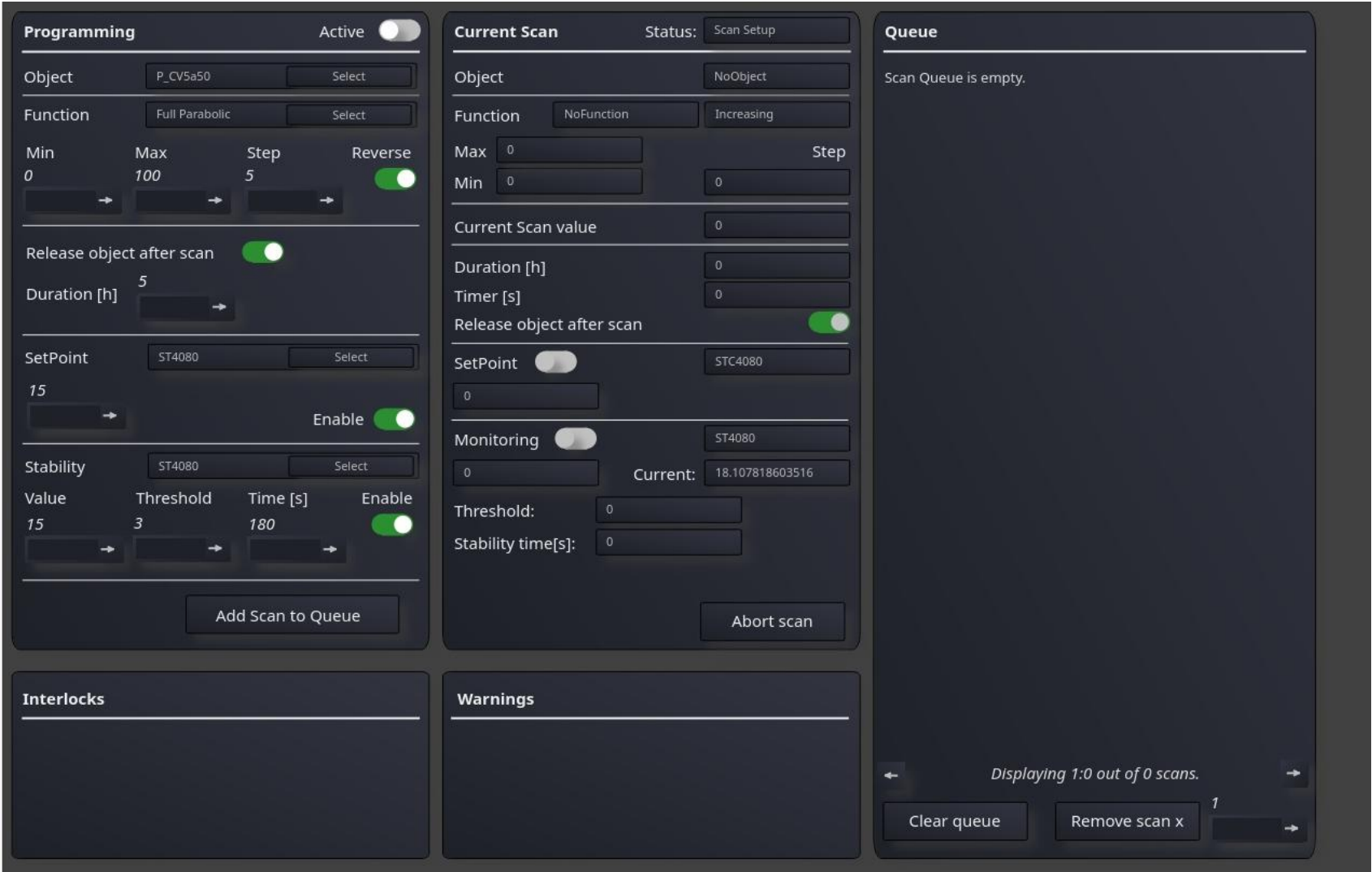
Accumulators:

- 3 x long accumulators } (8 controllers,
- 11 x short accumulators } 14 actuators,
- 160 IOs each)

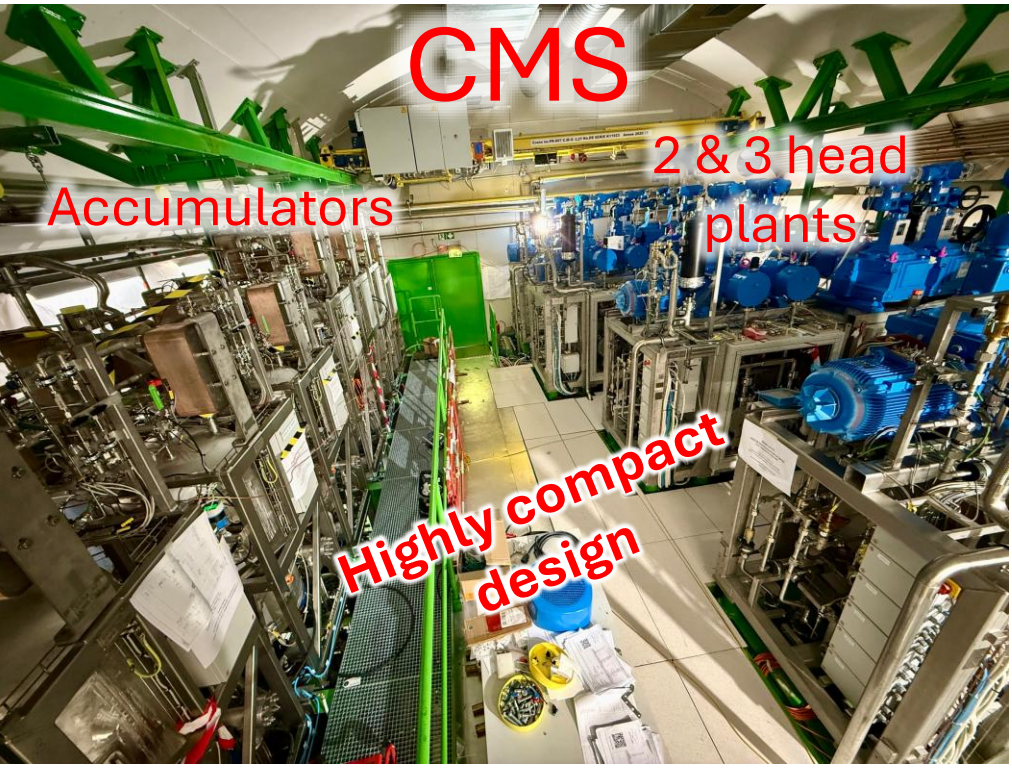
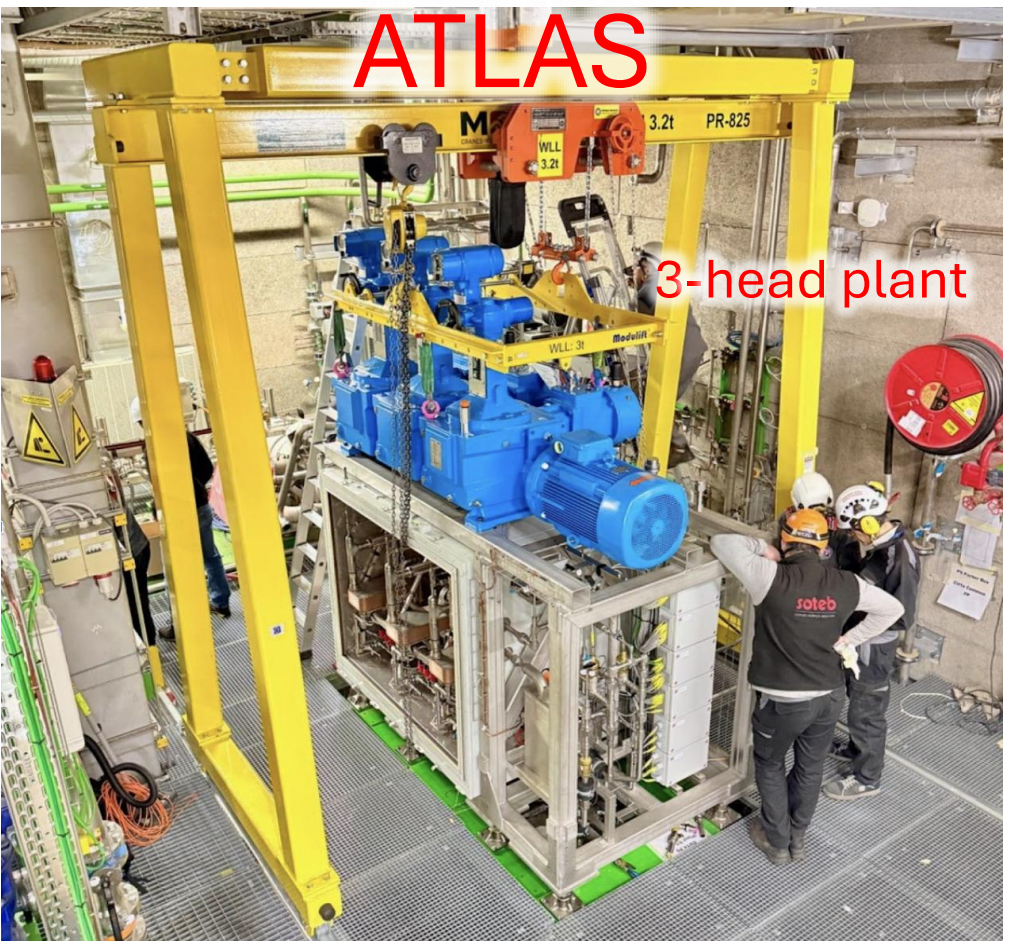
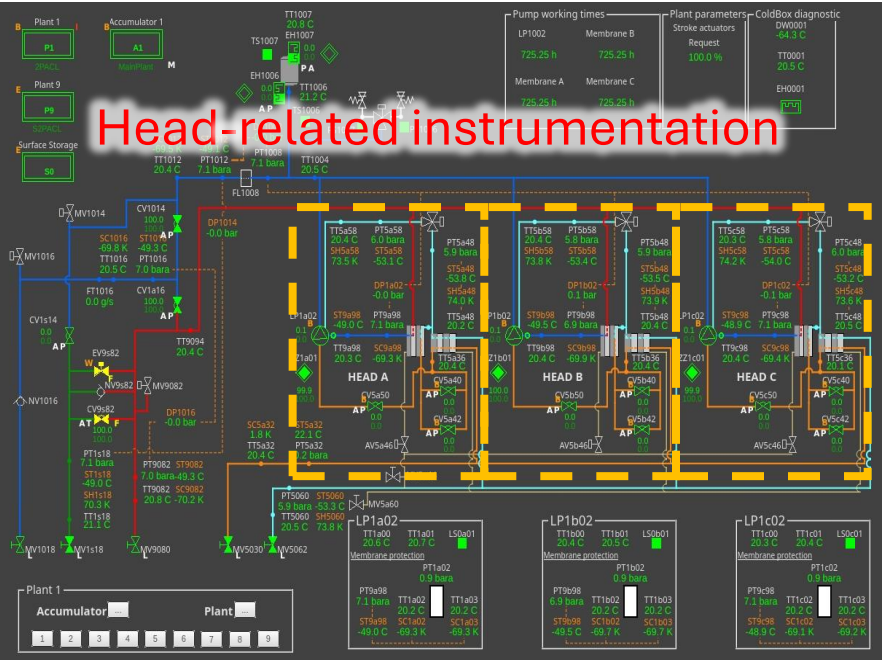
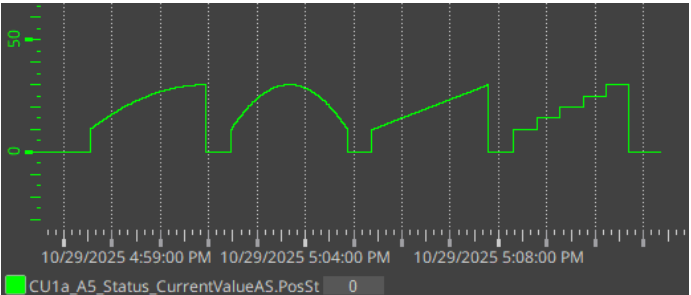
Manifolds:

- 14 systems unique per subdetector

Autoscan setup page



Autoscan curve options



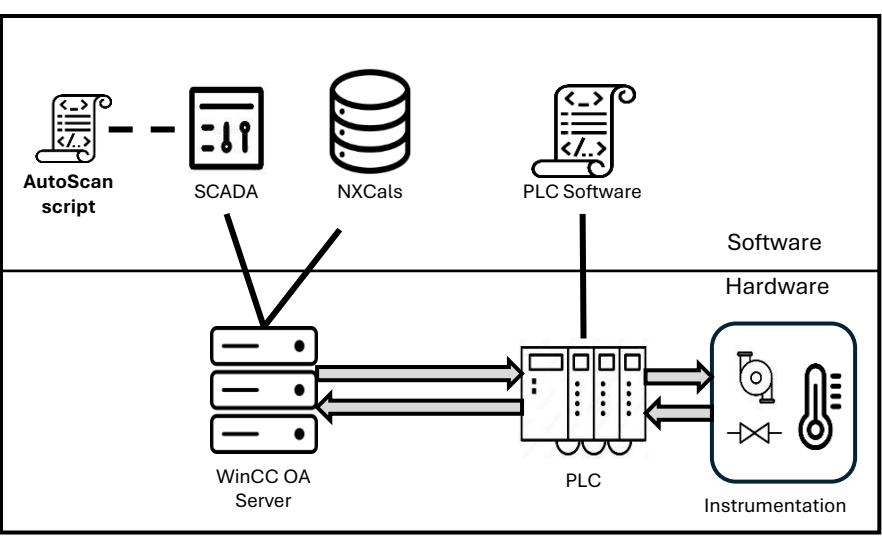
Requested Features:

- Automatic operation of different actuators
- Multiple predefined test scenarios such as steps, ramps and parabolic curves
- Customisable time frame for each scan
- Ability to queue up multiple tests

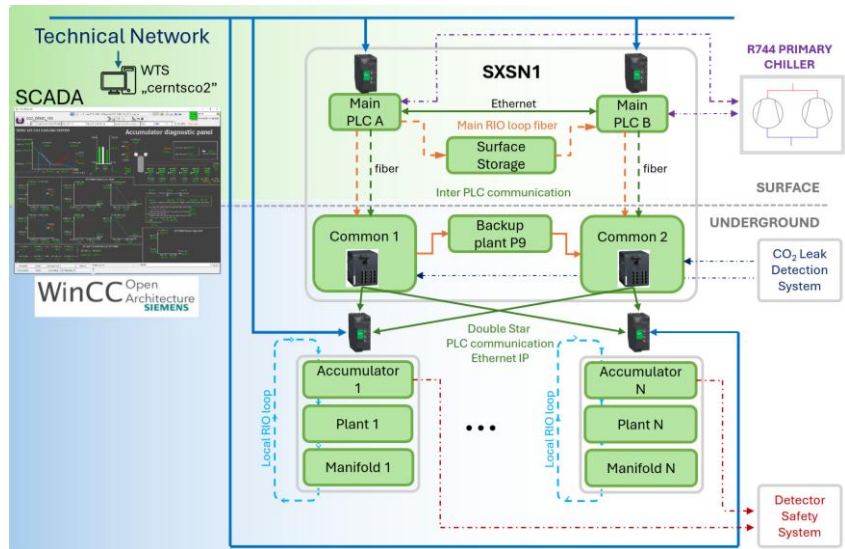
Development architecture:

- Autoscan is independent of PLC development
- Scripts are executed in the SCADA application
- Leverages UNICOS framework, simulating a manual operation from SCADA to the PLC
- PLC retains independent development strategy
- PLC remains the primary system for interlock in case of issues
- Access control handling follows the existing JCOP framework used in SCADA

Autoscan architecture



ATLAS & CMS control system architecture

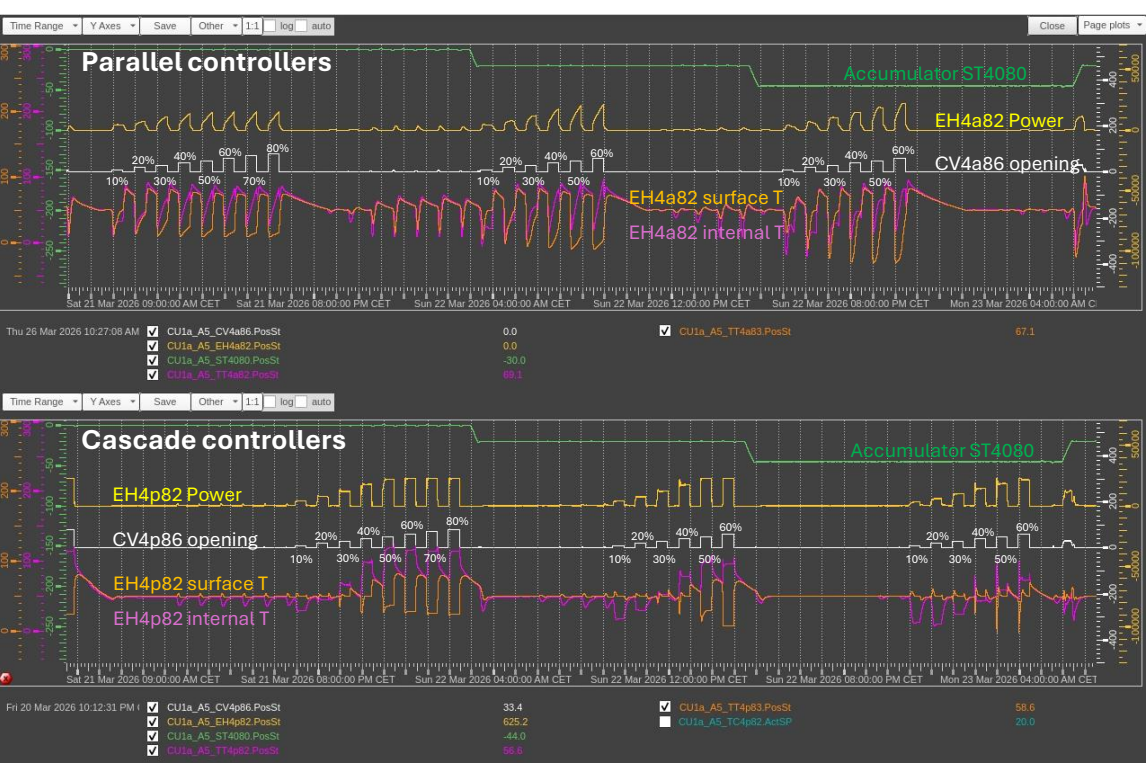
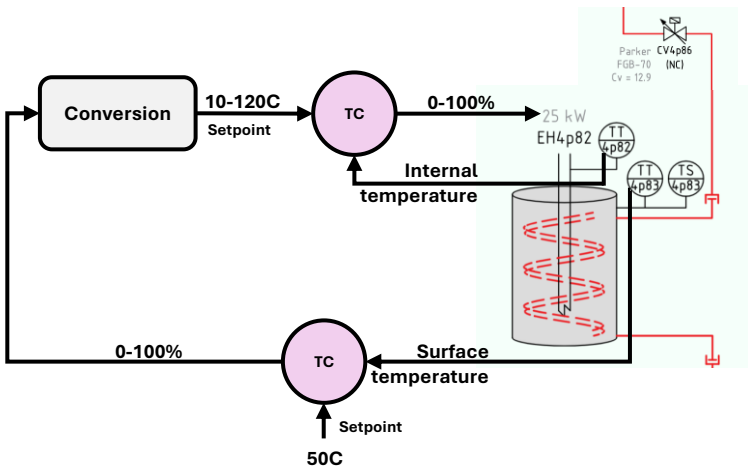


Current Results and Usage:

Each system has its own Autoscan script so systems operate independently during scans. These scripts are generated from one master file so updates are easily propagated to all systems. Scans are queued daily by the operation team and run through the full workday and over night or weekends. This leaves the operation team free to work on data processing or planning future test campaigns.

Example Test:

The accumulator heaters originally used 2 parallel controllers to regulate the temperature of the heater surface and the internal coil. A proposed solution was to replace this with a controller cascade where the output of the surface regulation feeds into the the setpoint of the internal regulation. These two solutions were able to be tested side by side to compare their behaviour by queuing several autoscans to apply steps in the output valve and see the controller response.



Future Development

- Add pump scan configuration
 - Needs to run a quick scan to find the boundaries of valve opening
 - Use these boundaries to launch a scan of the pump
- Create a dummy load reservation system to handle multiple scan requests with the dummy load which can connect to one system at a time